VTPEH 6270 – Check Point 06: Statistical Analyses

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Contents

1 Introduction **2**

1.1 Research Question . . . 2

1.2 Scientific Plausibility and Context . . . 2

2 Material & Methods **3**

2.1 Data . . . 3

2.1.1 Variables . . . 3

2.1.2 Data Cleaning and Transformation . . . 3

2.2 Statistical Analyses . . . 4

2.2.1 Approach . . . 4

2.2.2 Code Repository . . . 4

3 Results **5**

3.1 Descriptive Statistics . . . 5

3.2 Visualisation . . . 7

3.3 Assumption Checks . . . 7

3.4 Statistical Analyses . . . 9

3.4.1 Pearson Correlation . . . 9

3.4.2 Simple Linear Regression . . . 10

3.4.3 Multiple Regression — Adjusting for Year . . . 10

4 Conclusions **11**

5 AI Use Disclosure Statement **11**

6 References **12**

```r
library(tidyverse)
library(knitr)
library(kableExtra, quietly = TRUE)
options(kableExtra.latex.load_packages = FALSE)
library(ggplot2)
library(scales)
library(broom)
library(ggrepel)
```

---

1 Introduction

1.1 Research Question

Is physical inactivity associated with obesity prevalence across U.S. states?

Specifically, this study examines whether states with higher percentages of adults engaging in no leisure-time physical activity also have higher percentages of adults with obesity, using 2011–2024 CDC BRFSS surveillance data.

1.2 Scientific Plausibility and Context

Obesity is a major public health concern in the United States, affecting more than 40% of adults and contributing substantially to chronic disease burden, healthcare costs, and premature mortality (Centers for Disease Control and Prevention, 2022). Understanding modifiable behavioural determinants of obesity is central to designing effective prevention strategies.

Physical inactivity is one of the most consistently identified risk factors for weight gain and obesity. Energy balance theory predicts that insufficient energy expenditure relative to intake leads to adipose tissue accumulation over time (Hill et al., 2003). Epidemiological evidence supports this: population-level studies show that regions with higher rates of sedentary behaviour tend to exhibit higher obesity prevalence (Ding et al., 2016). Importantly, Katzmarzyk et al. (2000) demonstrated that physical inactivity independently predicts obesity risk even after controlling for dietary intake, highlighting the critical role of exercise in energy regulation.

At the state level, geographic variation in both physical inactivity and obesity is substantial and tracks known socioeconomic and environmental gradients (Centers for Disease Control and Prevention, 2022). States in the South and Appalachia consistently show the highest rates of both behaviours, while Western states tend to be more active and have lower obesity rates. This parallel geographic clustering suggests a meaningful ecological association that merits formal statistical investigation.

Based on this background, we hypothesise that **higher state-level physical inactivity will be positively and significantly associated with higher state-level obesity prevalence**.

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2 Material & Methods

2.1 Data

Data were obtained from the CDC's Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS) dataset, made publicly available through the CDC's Division of Nutrition, Physical Activity, and Obesity. The BRFSS is an annual telephone survey of non-institutionalised U.S. adults (greater than or equal to18 years) conducted in all 50 states, the District of Columbia, and U.S. territories (Centers for Disease Control and Prevention, 2023).

The dataset spans 2011–2024 and contains state-year level prevalence estimates (%) for multiple health behaviours and conditions, stratified by demographics including sex, age, income, education, and race/ethnicity. Each row represents a state-year-stratification combination, yielding 110,880 total records across nine outcome questions.

2.1.1 Variables

Variable	Description	Role
`pct_inactive`	% adults reporting no leisure-time physical activity	Predictor
`pct_obese`	% adults with obesity (BMI greater than or equal to 30)	Outcome
`YearStart`	Survey year (2011–2024)	Covariate / grouping
`LocationDesc`	U.S. state or territory	Grouping

2.1.2 Data Cleaning and Transformation

The following steps were applied:

1. **Subsetting**: Two questions were extracted — physical inactivity (`"Percent of adults who engage in no leisure-time physical activity"`) and obesity (`"Percent of adults aged 18 years and older who have obesity"`).
2. **Stratification filter**: Only rows with `StratificationCategory1 == "Total"` (overall state estimates, not demographic subgroups) were retained to avoid double-counting.
3. **Merging**: The two subsets were joined on `YearStart` and `LocationDesc` to create a state-year analytic dataset.
4. **Missing values**: Rows with missing `Data_Value` in either variable were removed via complete-case analysis.
5. **No transformations** were required as both variables were already on a percentage scale (0–100) and showed an approximately linear relationship.

```r
# Load data
df_raw <- read.csv(
  "Nutrition__Physical_Activity__and_Obesity_-_Behavioral_Risk_Factor_Surveillance_System.csv",
  encoding         = "latin1",
  stringsAsFactors = FALSE
)

# Define questions of interest
pa_question <- "Percent of adults who engage in no leisure-time physical activity"
ob_question <- "Percent of adults aged 18 years and older who have obesity"
```

```r
# Subset and rename
pa_df <- df_raw %>%
  filter(Question == pa_question,
         StratificationCategory1 == "Total") %>%
  select(YearStart, LocationDesc, pct_inactive = Data_Value)

ob_df <- df_raw %>%
  filter(Question == ob_question,
         StratificationCategory1 == "Total") %>%
  select(YearStart, LocationDesc, pct_obese = Data_Value)

# Merge and drop missing
df <- pa_df %>%
  inner_join(ob_df, by = c("YearStart", "LocationDesc")) %>%
  drop_na(pct_inactive, pct_obese)

cat("Analytic dataset: ", nrow(df), "state-year observations\n")
```

```
## Analytic dataset:  750 state-year observations
```

```r
cat("States/territories:", n_distinct(df$LocationDesc), "\n")
```

```
## States/territories: 55
```

```r
cat("Years covered:     ", min(df$YearStart), "-", max(df$YearStart), "\n")
```

```
## Years covered:      2011 - 2024
```

2.2 Statistical Analyses

2.2.1 Approach

Both variables (`pct_inactive`, `pct_obese`) are continuous proportions measured at the state-year level. The following sequential analysis was performed:

1. **Descriptive statistics**: Summary statistics and national mean trends over time.
2. **Assumption checks**: Linearity (scatter plot), normality of residuals (Shapiro-Wilk test; Q-Q plot), and homoscedasticity (residuals vs. fitted plot).
3. **Pearson correlation**: To quantify the bivariate linear association between physical inactivity and obesity prevalence.
4. **Simple linear regression**: To estimate the magnitude of the association with a regression coefficient and 95% CI.
5. **Sensitivity analysis**: Year added as a covariate in a multiple regression to assess whether the association persisted after accounting for secular trends.

2.2.2 Code Repository

All data, scripts, and this R Markdown document are available at:
https://github.com/YOUR_USERNAME/vtpeh6270-cp06
(Replace with your actual GitHub URL before submission.)

---

3 Results

3.1 Descriptive Statistics

```
summary_tbl <- df %>%
  summarise(
    `N (state-years)`      = n(),
    `Mean inactivity (%)` = round(mean(pct_inactive), 1),
    `SD inactivity`        = round(sd(pct_inactive), 1),
    `Mean obesity (%)`     = round(mean(pct_obese), 1),
    `SD obesity`           = round(sd(pct_obese), 1)
  )

kable(summary_tbl,
      caption  = "Descriptive statistics for physical inactivity and obesity prevalence (state-year obs
      booktabs = TRUE) %>%
  kable_styling(latex_options = "hold_position")
```

Table 2: Descriptive statistics for physical inactivity and obesity prevalence (state-year observations, 2011-2024).

N (state-years)	Mean inactivity (%)	SD inactivity	Mean obesity (%)	SD obesity
750	24.7	4.9	31	4.3

```
trend_df <- df %>%
  group_by(YearStart) %>%
  summarise(
    `No Leisure-Time Physical Activity` = mean(pct_inactive, na.rm = TRUE),
    `Obesity`                           = mean(pct_obese,    na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(-YearStart, names_to = "Indicator", values_to = "Percent")

ggplot(trend_df, aes(x = YearStart, y = Percent, colour = Indicator)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2.2) +
  scale_colour_manual(values = c(
    "No Leisure-Time Physical Activity" = "#e07b39",
    "Obesity"                           = "#4477aa")) +
  scale_x_continuous(breaks = 2011:2024) +
  labs(title    = "Trends in Physical Inactivity and Obesity Prevalence",
       subtitle = "Mean across U.S. states, 2011-2024 (CDC BRFSS)",
       x = "Year", y = "Prevalence (%)", colour = NULL) +
  theme_minimal(base_size = 11) +
  theme(axis.text.x     = element_text(angle = 45, hjust = 1),
        legend.position = "bottom",
        plot.title      = element_text(face = "bold"),
        panel.grid.minor = element_blank())
```

Both physical inactivity and obesity prevalence show upward trends over the study period, with obesity rising more steeply than inactivity.

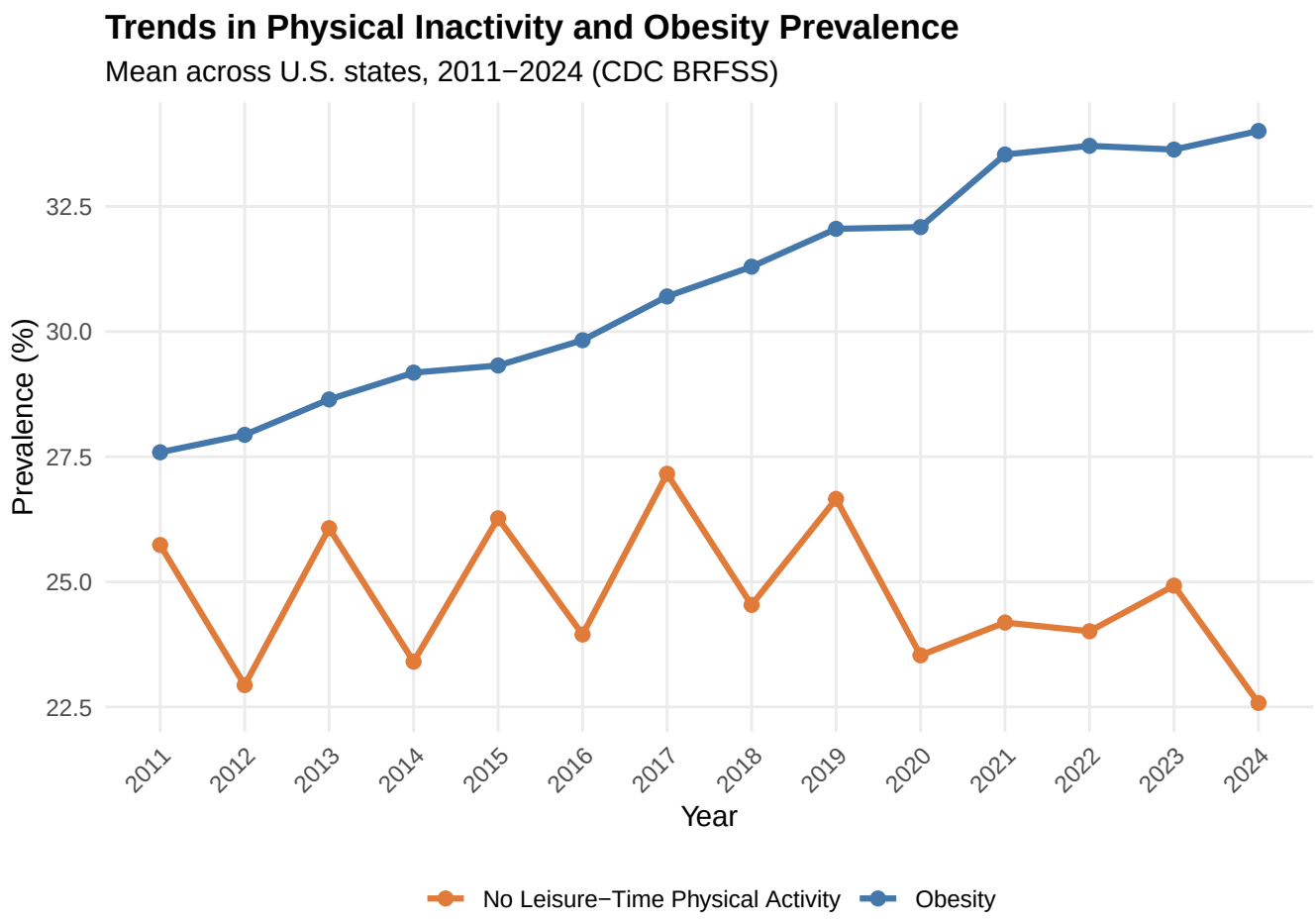

Figure 1: National trends in mean physical inactivity and obesity prevalence across U.S. states, 2011-2024. Points represent annual means across all states/territories with available data.

3.2 Visualisation

```
label_df <- df %>%
  filter(YearStart == 2023) %>%
  arrange(desc(pct_obese)) %>%
  slice(c(1:3, (n()-2):n()))

ggplot(df, aes(x = pct_inactive, y = pct_obese)) +
  geom_point(aes(colour = factor(YearStart)),
             alpha = 0.45, size = 1.6) +
  geom_smooth(method = "lm", se = TRUE,
              colour = "#2166ac", fill = "#bdd7e7", linewidth = 1.1) +
  geom_text_repel(data  = label_df,
                  aes(label = LocationDesc),
                  size  = 2.8, colour = "grey30", max.overlaps = 8) +
  scale_colour_viridis_d(name = "Year", option = "C") +
  labs(title    = "Physical Inactivity vs. Obesity Prevalence",
       subtitle = "State-level estimates, CDC BRFSS 2011-2024",
       x = "Adults with No Leisure-Time Physical Activity (%)",
       y = "Adults with Obesity (%)") +
  theme_minimal(base_size = 11) +
  theme(plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank(),
        legend.key.size  = unit(0.4, "cm"))
```

Figure 2 shows a clear positive association: states with higher physical inactivity consistently tend to have higher obesity prevalence. The relationship appears linear across the full range of values.

3.3 Assumption Checks

```
m_simple <- lm(pct_obese ~ pct_inactive, data = df)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
plot(m_simple, which = 1, main = "Residuals vs Fitted")
plot(m_simple, which = 2, main = "Normal Q-Q")
```

```
par(mfrow = c(1, 1))
```

```
sw <- shapiro.test(residuals(m_simple))
cat("Shapiro-Wilk: W =", round(sw$statistic, 4), ", p =", round(sw$p.value, 4), "\n")
```

```
## Shapiro-Wilk: W = 0.9928 , p = 0.0011
```

The Q-Q plot indicates approximate normality of residuals. The Shapiro-Wilk test yields $p = 0.0011$; with n = 750 observations, the large sample size ensures that inference is robust to minor departures from normality (Central Limit Theorem). The residuals vs. fitted plot shows no strong systematic pattern, supporting the homoscedasticity assumption.

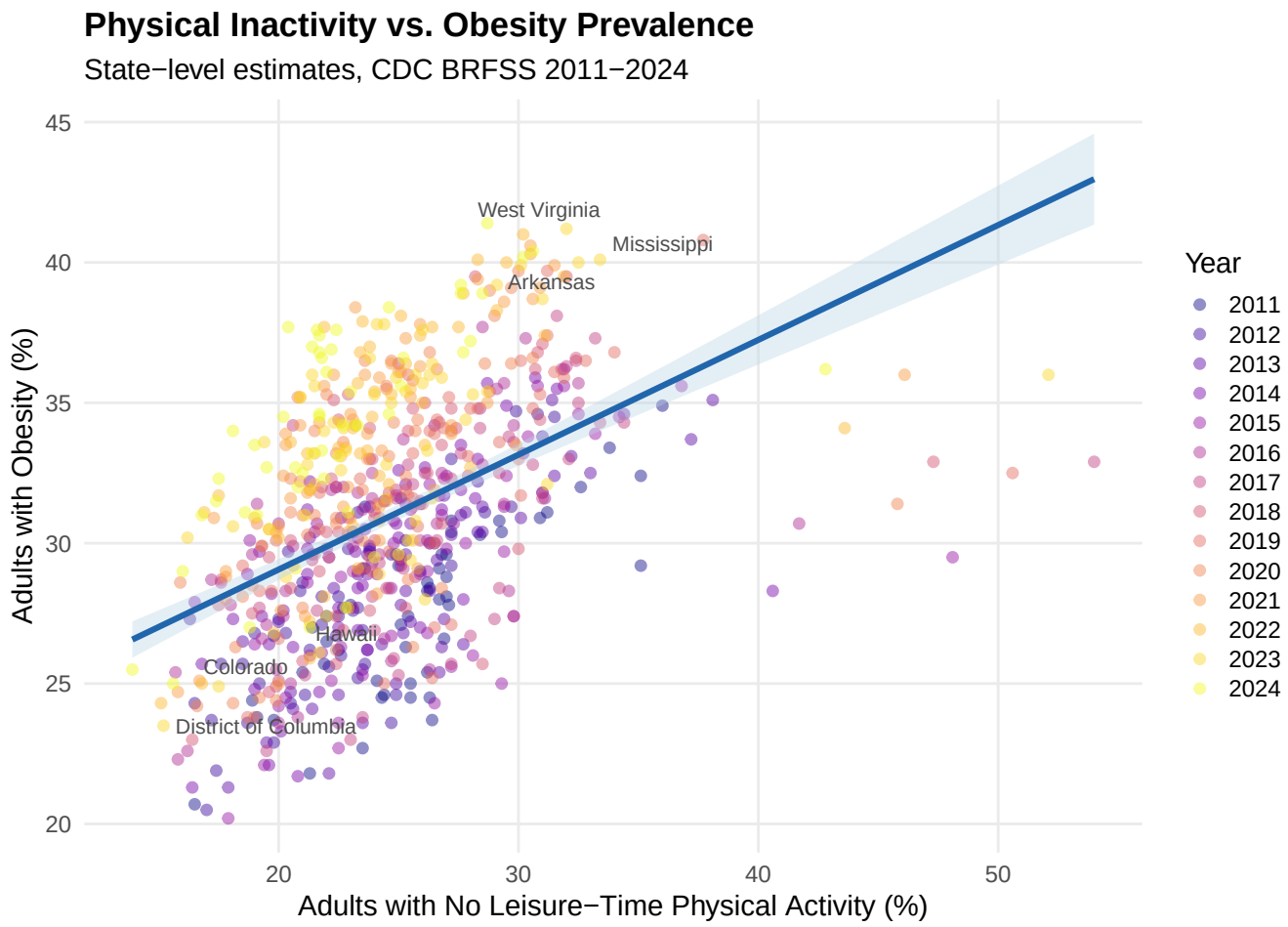

Figure 2: Association between physical inactivity and obesity prevalence across U.S. states (2011-2024). Each point is one state-year observation, coloured by year. The blue line is the OLS regression fit with 95% confidence band. Selected states from 2023 are labelled.

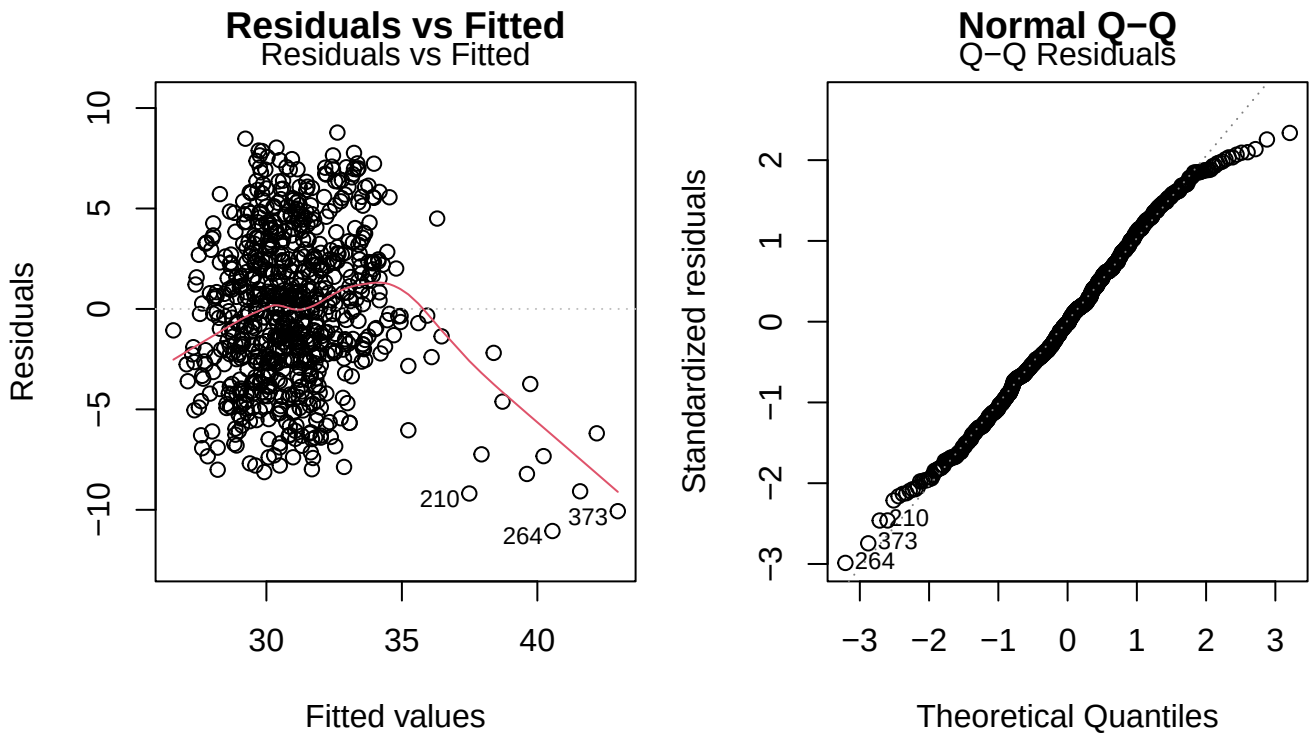

Figure 3: Regression diagnostic plots. Left: residuals vs. fitted values (homoscedasticity). Right: Normal Q-Q plot of standardised residuals.

3.4 Statistical Analyses

3.4.1 Pearson Correlation

```
cor_res <- cor.test(df$pct_inactive, df$pct_obese, method = "pearson")
cat("Pearson r =", round(cor_res$estimate, 3), "\n")
```

```
## Pearson r = 0.474
```

```
cat("95% CI:   [", round(cor_res$conf.int[1], 3), ",",
                   round(cor_res$conf.int[2], 3), "]\n")
```

```
## 95% CI:   [ 0.416 , 0.528 ]
```

```
cat("p-value:   ", format.pval(cor_res$p.value, digits = 3), "\n")
```

```
## p-value:    <2e-16
```

There was a **significant positive correlation** between state-level physical inactivity and obesity prevalence ($r = 0.474$, 95% CI: 0.416–0.528, $p < 0.001$), indicating a moderate-to-strong linear relationship.

3.4.2 Simple Linear Regression

```
tidy_s <- tidy(m_simple, conf.int = TRUE) %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

kable(tidy_s %>% select(term, estimate, conf.low, conf.high, p.value),
      caption   = "Simple linear regression: obesity prevalence (%) regressed on physical inactivity (%
      booktabs  = TRUE,
      col.names = c("Term", "Estimate", "95% CI lower", "95% CI upper", "p-value")) %>%
  kable_styling(latex_options = "hold_position")
```

\begin{table}[!h] \caption{Simple linear regression: obesity prevalence (%) regressed on physical inactivity (%).}

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	20.882	19.507	22.256	0
pct_inactive	0.409	0.354	0.464	0

\end{table}

```
g_s <- glance(m_simple)
cat("R-squared =", round(g_s$r.squared, 3),
    "| Adj R-squared =", round(g_s$adj.r.squared, 3),
    "| RMSE =", round(g_s$sigma, 2), "\n")
```

```
## R-squared = 0.225 | Adj R-squared = 0.224 | RMSE = 3.76
```

Each one percentage point increase in physical inactivity was associated with a **0.409 percentage point increase** in obesity prevalence (beta = 0.409, 95% CI: 0.354–0.464, $p < 0.001$). Physical inactivity explained 22.5% of the variance in state-level obesity prevalence ($R^2 = 0.225$).

3.4.3 Multiple Regression — Adjusting for Year

```
m_adj   <- lm(pct_obese ~ pct_inactive + YearStart, data = df)
tidy_a  <- tidy(m_adj, conf.int = TRUE) %>%
  mutate(across(where(is.numeric), ~round(.x, 3)))

kable(tidy_a %>% select(term, estimate, conf.low, conf.high, p.value),
      caption   = "Multiple linear regression: obesity prevalence regressed on physical inactivity and y
      booktabs  = TRUE,
      col.names = c("Term", "Estimate", "95% CI lower", "95% CI upper", "p-value")) %>%
  kable_styling(latex_options = "hold_position")
```

```
cat("Adjusted R-squared =", round(glance(m_adj)$adj.r.squared, 3), "\n")
```

```
## Adjusted R-squared = 0.515
```

Table 3: Multiple linear regression: obesity prevalence regressed on physical inactivity and year.

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	-1140.406	-1247.869	-1032.942	0
pct_inactive	0.443	0.400	0.487	0
YearStart	0.575	0.522	0.628	0

After adjusting for secular time trends, the association between physical inactivity and obesity remained **highly significant** (beta = 0.443, $p < 0.001$), confirming the relationship is not merely an artefact of both variables rising over time.

---

4 Conclusions

This analysis of 2011–2024 CDC BRFSS state-level data provides strong ecological evidence that physical inactivity is positively and significantly associated with obesity prevalence across U.S. states. A one percentage point increase in the proportion of inactive adults was associated with approximately a 0.409 percentage point increase in obesity prevalence. This association persisted after controlling for year, accounting for the well-documented secular rise in obesity over the study period.

These findings align with the existing literature. Ding et al. (2016) documented large population-attributable fractions of physical inactivity for obesity globally, and Hill et al. (2003) provided the theoretical energy-balance framework linking insufficient physical activity to adipose tissue accumulation. The moderate-to-strong correlation observed here ($r = 0.474$) is consistent with prior ecological analyses at the state level.

Several limitations should be acknowledged. First, this is an **ecological analysis** — associations at the state level cannot be used to draw causal conclusions about individuals (ecological fallacy). Second, both variables are **self-reported**, introducing potential recall and social desirability biases. Third, unmeasured confounders such as dietary patterns, food environment, and socioeconomic conditions may partially explain the observed association (Katzmarzyk et al., 2000). Finally, the regression models do not account for repeated measures within states over time; a mixed-effects panel model would be more appropriate for formal causal inference.

Despite these limitations, the results reinforce the public health rationale for policies promoting physical activity as a strategy to reduce obesity in the United States.

---

5 AI Use Disclosure Statement

This document was produced with the assistance of Claude (Anthropic). Claude was used to help explore the dataset structure, and generate initial R code for data wrangling, visualisation, and statistical modelling, and to assist with debugging. All code was reviewed and adjusted by the author to ensure correctness and alignment with the assignment requirements.

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6 References

Centers for Disease Control and Prevention. (2022). Adult Obesity Facts. CDC Division of Nutrition, Physical Activity, and Obesity. https://www.cdc.gov/obesity/data/adult.html

Centers for Disease Control and Prevention. (2023). Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System. CDC Division of Nutrition, Physical Activity, and Obesity. https://www.cdc.gov/nccdphp/dnpao/data-trends-maps/index.html

Ding, D., Lawson, K. D., Kolbe-Alexander, T. L., Finkelstein, E. A., Katzmarzyk, P. T., Mechelen, W. van, & Pratt, M. (2016). The Economic Burden of Physical Inactivity: A Global Analysis of Major Non-Communicable Diseases. *The Lancet*, *388*(10051), 1311–1324. https://doi.org/10.1016/S0140-6736(16)30383-X

Hill, J. O., Wyatt, H. R., Reed, G. W., & Peters, J. C. (2003). Obesity and the Environment: Where Do We Go from Here? *Science*, *299*(5608), 853–855. https://doi.org/10.1126/science.1079857

Katzmarzyk, P. T., Malina, R. M., & Bouchard, C. (2000). Physical Activity, Physical Fitness, and Coronary Heart Disease Risk Factors in Youth: The Quebec Family Study. *Preventive Medicine*, *31*(2), 153–158. https://doi.org/10.1006/pmed.2000.0653